

# Teaching Dossier

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Computer Science / Software Engineering

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## 1. Teaching Philosophy and Approach

My teaching is rigorous, inclusive, and professionally relevant. I design learning experiences so students connect computing and software engineering theory with implementation, justify design decisions using evidence, and build communication, collaboration, and ethical reasoning skills required in professional practice.

I use outcomes-based course design, active learning, and structured formative feedback. Classes typically progress from foundational concepts through guided labs and practical assignments to integrative projects. Students are expected not only to produce working solutions, but also to justify architectural and analytical decisions, communicate technical rationale, and engage in collaborative problem solving.

## 2. Teaching Experience and Course Portfolio

My portfolio spans university and polytechnic teaching, with lecture and laboratory responsibilities across programming, systems, networking, cloud computing, software development, blockchain technologies, and project-oriented computing.

### Current and recent teaching roles

- **Instructor**, School of Computing and Digital Innovation, Saskatchewan Polytechnic (Regina Campus), 2024–Present.
- **Sessional Lecturer**, Department of Computer Science, University of Saskatchewan — CMPT 145.3 Principles of Computer Science (Spring/Summer 2025; sections MT1 and MT2).
- **Lecturer**, CSE Discipline, Khulna University, Bangladesh (2013–2017 active; 2017–2024 on study leave).
- **Part-time Lecturer**, North Western University, Khulna — Computer Science courses for the BA program (2014–2016).
- **Instructor (Part-time)**, Higher Secondary Teachers' Training Institute (HSTTI), Khulna — ICT courses for high school teachers (2014–2016).
- **Teaching Assistant / Student Assistantship**, University of Saskatchewan (2017–2024), including CMPT 214, 145, 370, 280, 270, 141, and Database Management Systems.

**Representative courses taught** (from institutional teaching records)

- CMPT 145: Principles of Computer Science
- COOS 190 / 294 / 295: Server Administration; Cloud Infrastructure Administration; Systems Administration 2
- COET 295: Emerging Technologies; CNET 184: Data Communications and Networking; CWEB 280: Internet Programming and Web Applications 2
- Cloud Computing & Blockchain suite (CCMP 600–606), DEVP 600 (DevOps), PROJ 611, and related offerings
- Khulna University: Structured Programming; Software Development Project; Advanced Programming Laboratory; Software Engineering and Information System

I have also designed and/or rewritten course and lab materials for most Saskatchewan Polytechnic courses taught. This breadth supports software engineering core instruction and interdisciplinary contributions across computing programs.

### 3. Evidence of Teaching Effectiveness

Evidence in this package includes institutional teaching records, sample course outlines, and student evaluation materials. Across offerings, student outcomes and feedback reflect clarity of explanation, responsiveness to learner needs, and practical relevance of assignments and labs.

Course evaluation evidence includes an end-of-course SLEQ report for CMPT 145 (University of Saskatchewan, Spring/Summer 2025). The offering was below the institutional response threshold for closed-ended scores; qualitative comments are included. Representative student feedback themes include clear and accessible explanations for early-stage learners, responsive communication and flexibility, and a strong commitment to student learning. I also treat engagement design in lower-year cohorts as an ongoing improvement priority.

I use teaching evidence formatively to improve instructional sequencing, assessment design, active-learning prompts, and feedback quality, strengthening engagement and measurable learning outcomes over time.

### 4. Sample Syllabi and Course Materials

Included sample outlines and learning artifacts illustrate outcomes-based design, coherent assessment structures, and progressive skill development through labs and projects.

**CNET 184** (Data Communications and Networking): approximately 60 contact hours; assessment mix of labs, quizzes, assignment, and exams. Learning outcomes include network models and properties, TCP/IP services, monitoring and troubleshooting, and network security mechanisms.

**CWEB 280** (Internet Programming / Web Applications 2): approximately 64 contact hours; assessment through assignments and projects. Outcomes include server-side web development, secure authentication, state management, SQL-backed data handling, RESTful patterns, and full-stack project delivery using contemporary web frameworks and API workflows.

### 5. Teaching Innovation and Continuous Improvement

I continuously refine course delivery through reflective practice and evidence-based updates. I incorporate contemporary engineering workflows—version control, testing pipelines, reproducible development environments, and iterative milestone feedback—where pedagogically appropriate, so students build both conceptual depth and practical readiness for capstone work and professional practice.

I also introduce structured team processes and authentic technical tasks that mirror real software and systems workflows, while adjusting rubrics and classroom activities based on learner performance patterns.

### 6. Equity, Diversity, Inclusion, and Accessibility (EDIA)

I design courses with transparent expectations, accessible learning resources, structured rubrics, timely formative feedback, and inclusive participation structures. I foster classroom environments where diverse perspectives are welcomed and where students from varied learning backgrounds can thrive while maintaining academic rigor.

My mentoring approach emphasizes belonging, fairness, and support. I integrate responsible technology discussions—including ethics, accessibility, and social impact—into computing and software engineering learning. Ongoing professional learning includes Reconciliation Education / Four Seasons of Reconciliation, which informs respectful classroom culture, inclusive examples, and accessible course structures.

## **7. Instructional Development and Professional Engagement**

I am committed to sustained growth as an educator through reflective practice, curriculum development, and formal teaching development programs, including:

- Learners, Instruction & Fundamentals of Teaching (LIFT), Saskatchewan Polytechnic — 90 hours (completed May 2026), with an instructional ePortfolio.
- New Instructor Orientation (NIO), Saskatchewan Polytechnic — 18 hours (January 2025).
- Selected Faculty Participant, China Henan (ZJTIE) Transnational Education Project, Saskatchewan Polytechnic (2026).
- CETL / IQAC pedagogy trainings, Khulna University (2014–2016).
- 14th Foundation Training for University Teachers (FTUT), Graduate Training Institute, Bangladesh Agricultural University — 150 hours (2014).

## **8. Mentorship and Student Support**

I mentor students in research and teaching contexts, including M.Sc. mentoring in software backporting and quality analysis, supervision of undergraduate summer research students, and guidance of teaching assistants while serving as Sessional Lecturer for CMPT 145. I am particularly interested in mentoring projects in software engineering methodologies, automated code analysis and backporting, and machine learning applications in software development.

## **Appendix A. Teaching Evidence**

Supporting teaching records and course materials. Additional certificates, the LIFT instructional ePortfolio, and related artifacts are available at <https://joydeba.github.io/teaching-materials/>.

- Course / CRN teaching records (Saskatchewan Polytechnic).
- SLEQ end-of-course report for CMPT 145.01 Principles of Computer Science (University of Saskatchewan, Spring/Summer 2025).
- Course outline: CNET 184 — Data Communications and Networking.
- Course outline: CWEB 280 — Internet Programming and Web Applications 2.

Faculty & Advisors   CRN Listing

CRN Listing

| All Terms     |                                             |                          |                  |        |                                                  |                                    |
|---------------|---------------------------------------------|--------------------------|------------------|--------|--------------------------------------------------|------------------------------------|
| Select Course |                                             |                          |                  |        | <input type="text" value="Search CRN and Term"/> |                                    |
| Subject       | Course Title                                | CRN                      | Enrollment Count | Status | Duration                                         | Term                               |
| Subject       | Course Title                                | CRN                      | Enrollment Count | Status | Duration                                         | Term                               |
| COOS 295, T02 | Systems Administration 2<br>(../courseList) | 1392<br>(../courseList)  | 24               | Active | 05/04/2026 - 06/12/2026                          | 2025/2026 Spring Semester (202504) |
| COOS 295, T03 | Systems Administration 2<br>(../courseList) | 1393<br>(../courseList)  | 25               | Active | 05/04/2026 - 06/12/2026                          | 2025/2026 Spring Semester (202504) |
| COET 295, T02 | Emerging Technologies<br>(../courseList)    | 1408<br>(../courseList)  | 22               | Active | 05/04/2026 - 06/12/2026                          | 2025/2026 Spring Semester (202504) |
| COET 295, T03 | Emerging Technologies<br>(../courseList)    | 1423<br>(../courseList)  | 25               | Active | 05/04/2026 - 06/12/2026                          | 2025/2026 Spring Semester (202504) |
| COOS 190, T02 | Server Administration<br>(../courseList)    | 12056<br>(../courseList) | 9                | Active | 01/05/2026 - 04/28/2026                          | 2025/2026 Winter Semester (202503) |
| COOS 190, T03 | Server Administration<br>(../courseList)    | 12068<br>(../courseList) | 26               | Active | 01/05/2026 - 04/28/2026                          | 2025/2026 Winter Semester (202503) |
| COOS 294, T04 | Cloud Infrastructure Admin                  | 12525<br>(../courseList) | 23               | Active | 01/05/2026 - 04/28/2026                          | 2025/2026 Winter Semester          |

|                  |                                                         |                                        |        |                            |  |                                             |
|------------------|---------------------------------------------------------|----------------------------------------|--------|----------------------------|--|---------------------------------------------|
|                  | (../courseList)                                         |                                        |        |                            |  | (202503)                                    |
| COOS 294,<br>T05 | Cloud<br>Infrastructure<br>Admin<br>(../courseList)     | 12531<br>(../courseList) <sup>25</sup> | Active | 01/05/2026 -<br>04/28/2026 |  | 2025/2026<br>Winter<br>Semester<br>(202503) |
| CPMG 290,<br>T02 | IT<br>Development<br>Project<br>Mgmt<br>(../courseList) | 12580<br>(../courseList) <sup>23</sup> | Active | 01/05/2026 -<br>04/28/2026 |  | 2025/2026<br>Winter<br>Semester<br>(202503) |
| CPMG 290,<br>T03 | IT<br>Development<br>Project<br>Mgmt<br>(../courseList) | 12581<br>(../courseList) <sup>23</sup> | Active | 01/05/2026 -<br>04/28/2026 |  | 2025/2026<br>Winter<br>Semester<br>(202503) |
| COSA 290,<br>T02 | IT<br>Development<br>Project 2<br>(../courseList)       | 12598<br>(../courseList) <sup>23</sup> | Active | 01/05/2026 -<br>04/28/2026 |  | 2025/2026<br>Winter<br>Semester<br>(202503) |
| COSA 290,<br>T03 | IT<br>Development<br>Project 2<br>(../courseList)       | 12599<br>(../courseList) <sup>23</sup> | Active | 01/05/2026 -<br>04/28/2026 |  | 2025/2026<br>Winter<br>Semester<br>(202503) |
| CNET 184,<br>T04 | Data<br>Communications/Network<br>1<br>(../courseList)  | 11440<br>(../courseList) <sup>10</sup> | Active | 08/28/2025 -<br>12/18/2025 |  | 2025/2026 Fall<br>Semester<br>(202502)      |
| CNET 184,<br>T05 | Data<br>Communications/Network<br>1<br>(../courseList)  | 11457<br>(../courseList) <sup>38</sup> | Active | 08/28/2025 -<br>12/18/2025 |  | 2025/2026 Fall<br>Semester<br>(202502)      |
| CWEB 280,<br>T02 | Internet<br>Program/Web<br>Apps 2<br>(../courseList)    | 11733<br>(../courseList) <sup>28</sup> | Active | 08/28/2025 -<br>12/18/2025 |  | 2025/2026 Fall<br>Semester<br>(202502)      |
| CWEB 280,<br>T03 | Internet<br>Program/Web<br>Apps 2<br>(../courseList)    | 11734<br>(../courseList) <sup>29</sup> | Active | 08/28/2025 -<br>12/18/2025 |  | 2025/2026 Fall<br>Semester<br>(202502)      |
|                  | Orchestration                                           |                                        |        |                            |  | 2024/2025                                   |

|                  |                                                         |                                        |        |                            |                                             |
|------------------|---------------------------------------------------------|----------------------------------------|--------|----------------------------|---------------------------------------------|
| CCMP 604,<br>T04 | Cloud<br>Resources<br>(../courseList)                   | 12261<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | Winter<br>Semester<br>(202403)              |
| CCMP 604,<br>L04 | Orchestration<br>Cloud<br>Resources<br>(../courseList)  | 12262<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | 2024/2025<br>Winter<br>Semester<br>(202403) |
| CCMP 605,<br>T04 | Cloud Data<br>Management<br>(../courseList)             | 12263<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | 2024/2025<br>Winter<br>Semester<br>(202403) |
| CCMP 605,<br>L04 | Cloud Data<br>Management<br>(../courseList)             | 12264<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | 2024/2025<br>Winter<br>Semester<br>(202403) |
| CCMP 606,<br>T04 | Integrated<br>Serv Smart<br>Contract<br>(../courseList) | 12265<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | 2024/2025<br>Winter<br>Semester<br>(202403) |
| CCMP 606,<br>L04 | Integrated<br>Serv Smart<br>Contract<br>(../courseList) | 12266<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | 2024/2025<br>Winter<br>Semester<br>(202403) |
| DEVP 600,<br>T04 | Business<br>Innovation/Dev<br>Ops<br>(../courseList)    | 12267<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | 2024/2025<br>Winter<br>Semester<br>(202403) |
| DEVP 600,<br>L04 | Business<br>Innovation/Dev<br>Ops<br>(../courseList)    | 12268<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | 2024/2025<br>Winter<br>Semester<br>(202403) |
| PROJ 611,<br>T01 | Adopt Cloud<br>Comp in<br>Industry<br>(../courseList)   | 12269<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | 2024/2025<br>Winter<br>Semester<br>(202403) |
| PROJ 611,<br>L04 | Adopt Cloud<br>Comp in<br>Industry<br>(../courseList)   | 12270<br>(../courseList) <sup>26</sup> | Active | 01/06/2025 -<br>04/29/2025 | 2024/2025<br>Winter<br>Semester<br>(202403) |

|                  |                                                       |                                       |        |                            |                                        |
|------------------|-------------------------------------------------------|---------------------------------------|--------|----------------------------|----------------------------------------|
| CCMP 600,<br>T01 | Fundamentals<br>Cloud<br>Computing<br>(../courseList) | 4397<br>(../courseList) <sup>26</sup> | Active | 08/29/2024 -<br>12/19/2024 | 2024/2025 Fall<br>Semester<br>(202402) |
| CCMP 601,<br>T01 | Fundamentals<br>of<br>Blockchain<br>(../courseList)   | 4398<br>(../courseList) <sup>26</sup> | Active | 08/29/2024 -<br>12/19/2024 | 2024/2025 Fall<br>Semester<br>(202402) |
| CCMP 602,<br>T01 | Provision/Security<br>Cloud Res.<br>(../courseList)   | 4399<br>(../courseList) <sup>26</sup> | Active | 08/29/2024 -<br>12/19/2024 | 2024/2025 Fall<br>Semester<br>(202402) |
| CCMP 603,<br>T01 | Intro to<br>Smart<br>Contracts<br>(../courseList)     | 4400<br>(../courseList) <sup>26</sup> | Active | 08/29/2024 -<br>12/19/2024 | 2024/2025 Fall<br>Semester<br>(202402) |
| TCOM 600,<br>T01 | Business<br>Technology<br>Comm<br>(../courseList)     | 4401<br>(../courseList) <sup>18</sup> | Active | 08/29/2024 -<br>12/19/2024 | 2024/2025 Fall<br>Semester<br>(202402) |
| CCMP 600,<br>L01 | Fundamentals<br>Cloud<br>Computing<br>(../courseList) | 4402<br>(../courseList) <sup>26</sup> | Active | 08/29/2024 -<br>12/19/2024 | 2024/2025 Fall<br>Semester<br>(202402) |
| CCMP 601,<br>L01 | Fundamentals<br>of<br>Blockchain<br>(../courseList)   | 4403<br>(../courseList) <sup>26</sup> | Active | 08/29/2024 -<br>12/19/2024 | 2024/2025 Fall<br>Semester<br>(202402) |
| CCMP 602,<br>L01 | Provision/Security<br>Cloud Res.<br>(../courseList)   | 4404<br>(../courseList) <sup>26</sup> | Active | 08/29/2024 -<br>12/19/2024 | 2024/2025 Fall<br>Semester<br>(202402) |
| CCMP 603,<br>L01 | Intro to<br>Smart<br>Contracts<br>(../courseList)     | 4405<br>(../courseList) <sup>26</sup> | Active | 08/29/2024 -<br>12/19/2024 | 2024/2025 Fall<br>Semester<br>(202402) |

Results found: 35

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**University of Saskatchewan End-of-course SLEQ – 202505/07 (Spr/Sum)**

**Below Response Threshold Individual End-of-course SLEQ  
Report – 202505/07 (Spr/Sum)**

**CMPT 145 01 Principles of Computer Science (CRN:41442)**

**Debasish Chakroborti**

**Role: Lecture (USask Core)**



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## USask Core Open-Ended Questions

**Please comment on the overall quality of the *instruction* provided by Debasish Chakroborti in this course.**

### Comment

The quality of instruction in this course was thoughtful and earnest, especially given that the professor is early in his teaching career. He demonstrated transparency about areas where he was still developing and made consistent efforts to engage students through class participation and prompt responses to questions. His explanations were clear and accessible, and he showed flexibility with accommodations when needed. While some aspects of delivery could be refined over time, his commitment to student learning and openness to feedback stood out positively. Overall, the instruction quality was above average for a new instructor, and I believe with continued experience, he will become an outstanding educator.

The lectures feels by the books, scripted almost. Everything day feels standard like an 'average' lecture. The in class quizzes are much more engaging than the online versions

**Please comment on any opportunities you had to develop and demonstrate subject specific skills in this course.**

### Comment

This course provided multiple opportunities to develop and demonstrate key subject-specific skills, including problem-solving, critical thinking, and application of core computer science concepts. Assignments and quizzes encouraged practical engagement with the material, fostering both individual understanding and collaborative learning. The mix of in-class activities and assessments helped reinforce theoretical knowledge while promoting analytical skills essential to the discipline. However, variability in class attendance sometimes limited the effectiveness of class participation components. Overall, the course structure supported skill development appropriate for an 100-level computer science class

**Please comment on the overall quality of your *learning experience* in this course.**

### Comment

This course made an effort to center participation and engagement, and the overall structure encouraged questions, student input, and flexibility. That said, a lack of engagement from many students significantly impacted the classroom environment. I believe that efforts to create a supportive, interactive space were sometimes undermined by student disengagement or unprofessional commentary outside class settings. This is not specific to this course, but seems to be a broader cultural issue in some lower-year CS cohorts, and is worth the department's attention.

# Course Outline

|                              |                                                                                                                                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Program</b>               | Computer Systems Technology                                                                                                                                                                                                                                               |
| <b>Course Code and Title</b> | CNET184 - Data Communications and Networking 1                                                                                                                                                                                                                            |
| <b>Course Description</b>    | In this course you will describe the major networking technologies and systems of modern networks, and be able to configure, manage and troubleshoot modern networks. This course presents content required in the objectives of the CompTIA Network+ certification exam. |
| <b>Prerequisite(s)</b>       | None                                                                                                                                                                                                                                                                      |
| <b>Corequisite(s)</b>        | None                                                                                                                                                                                                                                                                      |
| <b>Course Hours</b>          | 60 hours (4 hours/week x 15 weeks)                                                                                                                                                                                                                                        |
| <b>Credit Units</b>          | 4.0 units                                                                                                                                                                                                                                                                 |
| <b>Student Assessment</b>    | <p>Labs: 20%</p> <p>Quizzes: 20%</p> <p>Assignment: 10%</p> <p>Exams: 50% (2 exams, 25% each)</p>                                                                                                                                                                         |
| <b>Grade/Passing Grade</b>   | <ul style="list-style-type: none"><li>• Overall 50%</li><li>• Not yet developed</li></ul>                                                                                                                                                                                 |
| <b>PLAR Method</b>           |                                                                                                                                                                                                                                                                           |

**Learning Outcomes**

1. Explain network models.
2. Explain network properties.
3. Describe network implementations.
4. Configure the TCP/IP protocol .
5. Describe major TCP/IP services.
6. Monitor and troubleshoot networks.
7. Describe network attacks and security mechanisms.
8. Describe remote networking.
9. Identify network policies and procedures.

**Approvals**

Prepared/Updated by: Debasish Chakroborti

Date: Aug 2025

Approved by Program Head: Maxwell Christian

Date: Aug 2025

Approved by Academic Chair: Tanis Wilder

Date: June 2024



## Course Outline

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Course:</b>                             | <b>CWEB 280 – Internet Programming/Web Applications 2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    |
| <b>Course Description:</b>                 | You will receive instruction and practice in the development of server-side Web applications. You will learn how to write scripts that allow remote users to interface with databases existing on a World Wide Web server. You will become familiar with server-side programming to display website content dynamically as required.                                                                                                                                                                                                                                                                                      |                    |
| <b>Pre Requisites:</b>                     | <ul style="list-style-type: none"> <li>• COSC 183</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                    |
| <b>Co Requisites:</b>                      | <ul style="list-style-type: none"> <li>• CDBM 280 (concurrent)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    |
| <b>Course Hours:</b>                       | <ul style="list-style-type: none"> <li>• 64 (4 hours/week X 16 weeks)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                    |
| <b>Credit Units:</b>                       | <ul style="list-style-type: none"> <li>• 4</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                    |
| <b>Student Assessment</b>                  | <ul style="list-style-type: none"> <li>• Assignments: 20%</li> <li>• Midterm Projects (2): 40%</li> <li>• Final Project: 40%</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |
| <b>Grade/Passing Grade<br/>PLAR Method</b> | <ul style="list-style-type: none"> <li>• Overall 50%</li> <li>• Not yet determined</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                    |
| <b>Learning Outcomes:</b>                  | <ol style="list-style-type: none"> <li>1. Construct Web pages using basic server side language features.</li> <li>2. Construct secure Web pages using server side objects.</li> <li>3. Apply server side state management features.</li> <li>4. Manage database tables.</li> <li>5. Create objects to connect to a database server.</li> <li>6. Design web pages that retrieve and modify data stored in a SQL database.</li> <li>7. Use modern client/server communication methods to retrieve and modify Web page data.</li> <li>8. Develop a secure Web site using all the elements covered in this course.</li> </ol> |                    |
| <b>Prepared/Updated by:</b>                | <b>Debasish Chakroborti</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>August 2025</b> |
| <b>Approved by Program Head:</b>           | <b>Christian Maxwell</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>August 2025</b> |
| <b>Approved by Academic Chair:</b>         | <b>Osborne Gavin</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>August 2025</b> |

CWEB280

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**CWEB 280**

| Learning Outcomes/<br>Assessment Tools                                   | Learning Steps                                                                                                                                                                                                                                                                                                                                                                                                                                      | Learning<br>Activities/Resources                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Construct Web pages using basic server side language features.</b> | 1.1 Explain client side versus server side execution.<br>1.2 Utilize Node.js Express Framework with routing.<br>1.3 Create templates using Handlebars.js.<br>1.4 Utilize conditional statements and iterators in templates.<br>1.5 Create html forms to communicate with server.<br>1.6 Utilize 3rd party libraries to validate form inputs.<br>1.7 Utilize query strings to communicate with server.<br>1.8 Utilize Multer to handle file uploads. | 1.1 Lecture/demonstration/lab.<br>1.2 Lecture/demonstration/lab.<br>1.3 Lecture/demonstration/lab.<br>1.4 Lecture/demonstration/lab.<br>1.5 Lecture/demonstration/lab.<br>1.6 Lecture/demonstration/lab.<br>1.7 Lecture/demonstration/lab.<br>1.8 Lecture/demonstration/lab. |
| <b>Assessment Tools:</b><br>Project                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                              |
| <b>2. Construct secure Web pages using server side objects.</b>          | 2.1 Secure routes using JSON Web Tokens.<br>2.2 Discuss different methodologies for handling JWTs.<br>2.3 Utilize Passport.js to facilitate the use of JWTs<br>2.4 Discuss different Passport.js authentication strategies.                                                                                                                                                                                                                         | 2.1 Lecture/demonstration/lab.<br>2.2 Lecture/demonstration/lab.<br>2.3 Lecture/demonstration/lab.<br>2.4 Lecture/demonstration/lab.                                                                                                                                         |
| <b>Assessment Tools:</b><br>Project                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                              |

Commented [BE1]: We do class validation later in LO5 - no need to do it twice

| Learning Outcomes/<br>Assessment Tools                    | Learning Steps                                                                                                                                                                                                                                                                                                                                                      | Learning<br>Activities/Resources                                                                                                                                                                                                                                             |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture/demonstration/lab.                                |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                              |
| <b>3. Apply server side state management features.</b>    | 3.1 Describe what a cookie is and where it is appropriate to use a cookie.<br>3.2 Describe what a session variable is and where to use session variables.<br>3.3 Identify how Session variables are stored on a server.<br>3.4 Write web pages that set/read cookies.<br>3.5 Write web pages that set/read session variables                                        | 3.1 Lecture/demonstration/lab.<br>3.2 Lecture/demonstration/lab.<br>3.3 Lecture/demonstration/lab.<br>3.4 Lecture/demonstration/lab.<br>3.5 Lecture/demonstration/lab.                                                                                                       |
| <b>Assessment Tools:</b><br>Project                       |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                              |
| <b>4. Manage database tables.</b>                         | 4.1 Create database tables.<br>4.2 Alter database tables.<br>4.3 Delete database tables.<br>4.4 Create appropriate table relationships.<br>4.5 Insert data into database tables.<br>4.6 Update existing stored data from database tables.<br>4.7 Delete existing stored data from database tables.<br>4.8 Access information from tables using SQL query statements | 4.1 Lecture/demonstration/lab.<br>4.2 Lecture/demonstration/lab.<br>4.3 Lecture/demonstration/lab.<br>4.4 Lecture/demonstration/lab.<br>4.5 Lecture/demonstration/lab.<br>4.6 Lecture/demonstration/lab.<br>4.7 Lecture/demonstration/lab.<br>4.8 Lecture/demonstration/lab. |
| <b>Assessment Tools:</b><br>Project                       |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                              |
| <b>5. Create objects to connect to a database server.</b> | 5.1 Utilize TypeORM library to facilitate interactions with the database<br>5.2 Create TypeORM entities to represent database tables and records<br>5.3 Create controllers to handle RESTful requests to the server<br>5.4 Utilize 3 <sup>rd</sup> party libraries to validate data before storing in the database                                                  | 5.1 Lecture/demonstration/lab.<br>5.2 Lecture/demonstration/lab.<br>5.3 Lecture/demonstration/lab.<br>5.4 Lecture/demonstration/lab.                                                                                                                                         |
| <b>Assessment Tools:</b><br>Project                       |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                              |

| Learning Outcomes/<br>Assessment Tools                                                         | Learning Steps                                                                                                                                                                                                                           | Learning<br>Activities/Resources                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6. Design web pages that retrieve and modify data stored in a database.</b>                 | 6.1 Utilize the Vuejs frontend framework to facilitate creating web-based user interfaces<br>6.2 Utilize Bootstrap-Vue components to create responsive web-based user interfaces<br>6.3 Create mixins to communicate with the server api | 6.1 Lecture/demonstration/lab.<br>6.2 Lecture/demonstration/lab.<br>6.3 Lecture/demonstration/lab.                                                                                                                       |
| <b>Assessment Tools:</b><br>Project                                                            |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                          |
| <b>7. Use modern client/server communication methods to retrieve and modify Web page data.</b> | 7.1 Describe how Ajax (Asynchronous JavaScript and XML) works.<br>7.2 Create a web page that utilizes Ajax to send inputs to the server.<br>7.3 Create a web page that utilizes Ajax to fetch entries from a remote API.                 | 7.1 Lecture/demonstration/lab<br>7.2 Lecture/demonstration/lab<br>7.3 Lecture/demonstration/lab                                                                                                                          |
| <b>Assessment Tools:</b><br>Project                                                            |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                          |
| <b>8. Develop a secure Web site using all the elements covered in this course.</b>             | 8.1 Create a logical, user friendly, web application.<br>8.2 Create secure entity objects.<br>8.3 Develop individual web pages and forms.<br>8.4 Construct web page special effects.<br>8.5 Test the Web application.                    | 8.1 Create a full stack web application.<br>8.2 Create a full stack web application.<br>8.3 Create a full stack web application.<br>8.4 Create a full stack web application.<br>8.5 Create a full stack web application. |
| <b>Assessment Tools:</b><br>Project                                                            |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                          |